Miao Yang

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Educational Background

2022.09-Present **Tianjin University** **Ph. D**

- School of Materials Science and Engineering, Major in Materials science, Top 10%. (Expected Graduation Date: June 2026)
- Core Course: Physics of Metals, Frontal Issues in Materials Science and Engineering, Corrosion and Control.

2020.09-2022.06 **Tianjin University** **Master**

- School of Materials Science and Engineering, Major in Materials science, Top 10%.
- Core Course: Analysis Techniques in Materials Science, Thermodynamics of Nanomaterials, Transmission electron microscopes (TEM)-Basic, High Performance Computing and Applications.

2016.09-2020.06 **University of Science and Technology Beijing** **Bachelor**

- School of Materials Science and Engineering, Major in Materials Science and Engineering, Top 10%.
- Core Course: Fundamentals of Materials Science, Analysis Methods of Materials, Metallic Materials, Mechanical Behavior of Materials, Advanced Mathematics, Methods of Mathematical Physics, Engineering Mechanics.

Research Experience

2022-Present **Microstructure Control and Reliability Evaluation of Dissimilar Transient Liquid Phase Bonded Joints Between Nickel-Based Alloys and Austenitic Steels** **Independent Responsible Person**

- **Project Background:** Addressing the technical challenge of brittle phase formation at the joint interface between dissimilar materials (IN718 superalloy and 316LN austenitic steel) under high-temperature service conditions, which leads to poor reliability and short service life of the joint.
- **Project Execution:** Led the design and implementation of the project's technical roadmap. Designed and optimized key process windows for Transient Liquid Phase (TLP) bonding and subsequent Post-Bond Heat Treatment (PBHT). Through detailed characterization (SEM/EBSD) of interfacial micro-region phases, elemental diffusion behavior, the formation and elimination mechanisms of brittle phases were elucidated, establishing a reliable process-microstructure-property control model.
- **Project Results:** The developed process significantly enhanced the high-temperature creep life of the joint at 650°C from less than 1 hour to over 500 hours—an improvement exceeding 500-fold. This optimized the joint's microscopic morphology and resolved the core reliability issue for this material combination in high-temperature applications.

2020-2022 **Microstructural Evolution and Creep Fracture Mechanisms of Transient Liquid Phase Bonded Joints in High-Cr Ferritic Steels** **Core Contributor**

- **Project Background:** Focused on addressing the critical technical challenge of interfacial microstructure degradation in TLP bonded joints of high-Cr ferritic steels for ultra-supercritical thermal power applications, which leads to reduced service life under high-temperature creep conditions.
- **Project Execution:** Led the materials characterization and mechanistic analysis module. Integrated Thermo-Calc thermodynamic simulations with precise micro-regional composition/phase characterization (EDS/EBSD/TEM) to elucidate the non-equilibrium segregation of melting-point depressant elements (B, Si) and the precipitation behavior of secondary phases, revealing their dominant role in creep crack initiation and propagation mechanisms.
- **Project Results:** Demonstrated that the distribution of melting-point depressant elements is the critical factor determining joint creep life. The process optimization based on this mechanism achieved a 120% extension in creep rupture time at 650°C, effectively resolving the long-term stability challenges of this material system under high-temperature service conditions.

Research Publications

Published 5 SCI papers; 1 manuscript under review at Acta Materialia:

- **M. Yang**, T. Zhang, R. Ding, et al. Improving Creep Properties of IN718/316LN Transient Liquid Phase Bonding Joints by Controlling Evolution of Precipitate Phases. **Acta Materialia**, 2025, 121356.
- **M. Yang**, Y. Wang, R. Ding, et al. Precipitates evolution and fracture mechanism of the iso-thermally solidified TLP bonding joints between 316LN stainless steel and IN718 Ni-based alloy. **Materials Science and Engineering: A**, 2023, 145440.

Research Publications (continued)

- **M. Yang**, R. Ding, Q. Guo, et al. Kinetics of initial diffusion solidification during continuous heat-ing in term of IN718 superalloy / 316LN steel TLP bonding system. **Materialia**, 2024,102097s.
- **M. Yang**, B. Zhang, R. Ding, et al. Improving Tensile Properties of Dissimilar TLP Bonding Joints of IN718 Nickel-Based Superalloy/316LN Austenitic Steel by Long-Term Post-bonded Homogenization Treatment. **Metallurgical and Materials Transactions A**, 2024, 1958–1975.
- **M. Yang**, T. Zhang, R. Ding, et al. Microstructure Evolution of Athermal Solidification Zone and Its Effect on Mechanical Properties in Dissimilar Transient Liquid Phase Bonded Joint of IN718 Ni-Based Alloy/316LN Steel. **Journal of Materials Engineering and Performance**, 2023, 724–740.

Personal Skills

- ***Computational Materials Science:*** **Modeling** and **programming** for **DFT** calculations using **Materials Studio** and **VASP**
- ***Programming Skills:*** Developing codes for **MD** simulations using **LAMMPS**, and creating custom scripts with Python or Matlab
- ***Microstructural Characterization:*** Experienced with JSM 7800F thermal field emission SEM, JEM F200 TEM, S-4800 cold field emission SEM, and JXA iHP200F EPMA
- ***Software Proficiency:*** Word, PPT, LaTeX, VS Code, PyCharm, Matlab
- ***Language:*** English（fluent in speaking and writing）,Chinese(native),
- ***Additional Skills:*** Sample preparation (grinding, polishing, etching), mechanical testing (conventional/high-temperature tensile testing, creep testing)

Conference Attendance

- 2025.07 Poster presentation at China Materials Conference (2025) in Xiamen, Fujian Province, China
- 2023.06 Poster presentation at Far East NDT Forum (2023) in Tianjin, China

Honors & Awards

- 2025: National Scholarship for Ph.D. Students
- 2017, 2018, 2019: Outstanding Student of University of Science and Technology Beijing
- 2019: Third Prize in "Cradle Cup" Academic Science and Technology Competition
- 2020: Outstanding Graduate of University of Science and Technology Beijing
- 2016, 2017, 2018, 2019: People's Scholarship of University of Science and Technology Beijing
- 2020, 2021, 2022, 2023, 2024, 2025: Academic Excellence Scholarship of Tianjin University
- 2024: Outstanding Student of Tianjin University

References:

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2. **Prof. Chenxi Liu (Ph. D advisor)**
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